SAMSUNG

SentimentGuard

Real-time Sentiment Classification of Live Chat Comments

Team 6

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Big Data Course

Together for Tomorrow!

Enabling People

Education for Future Generations

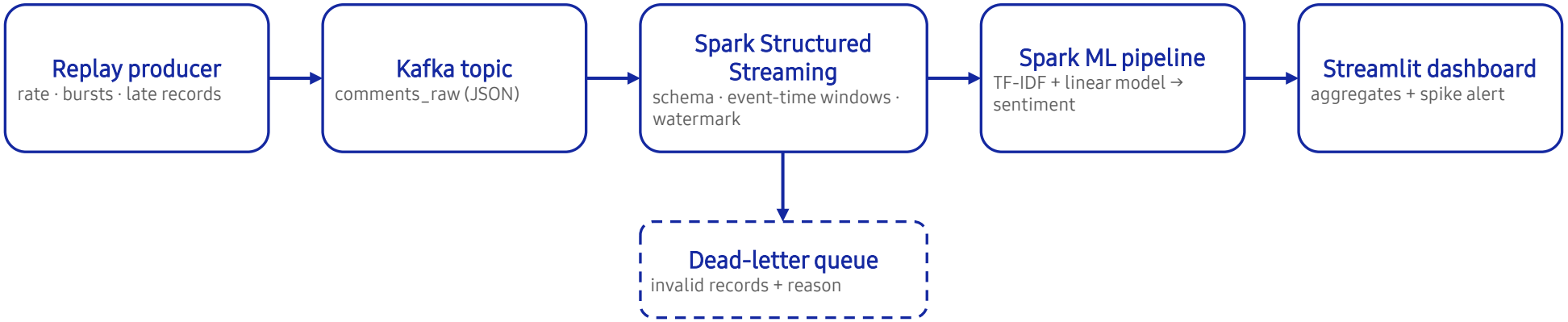
Problem and Technology for Good

- Live chat never stops — thousands of comments arrive far faster than one person can read.
- Harmful spikes (harassment, hate, spam) are noticed only after the damage is done; manual moderation alone cannot keep up.
- Goal: detect a negative spike within seconds and warn the streamer while the session is still live.
- Who benefits: solo and small streamers, including educational and public-service channels.
- Technology for Good: protects community safety and creator mental health — self-hosted, no chat data leaves the machine.

Data and Architecture

- Data: public IMDb reviews (CC0, about 1.0 million); 3 classes from 1–10 ratings.
- The stream is simulated: labelled live chat is unavailable, so we replay the data.
- Labels never enter the stream — stored separately, joined back by comment ID.
- Event time + watermark handle bursts; bad records go to a dead-letter queue.

Simulated live stream — the dataset is replayed through Kafka (not a real livestream):



Labels stay outside the stream: stored separately and joined by comment ID only for evaluation.

Expected Outcome and Novelty

- Near-real-time classification into negative / neutral / positive, with end-to-end latency reported (p50 / p95 / p99).
- Model quality reported as macro-F1 with per-class precision / recall / F1 — never accuracy alone.
- Novelty: detecting comment spikes, not just labelling comments — and one shared cleaning path for training and streaming, so no train/serve drift.
- Honest limits: rating-derived labels, domain shift (reviews are not chat), sarcasm; the stream is a replay — not a real livestream.
- Goal: a working prototype at TRL 5, validated in a simulated environment, released with reproducible scripts.

Roles and Schedule

- Ký — team lead, integration and testing; Bằng — Kafka ingestion and replay producer; Dẩm — streaming job, watermark, fault tolerance.
- Hạnh — data preparation and the Spark ML model; Nguyên — SQL queries, figures and the Streamlit dashboard with alerts.
- Phase 1 (Day 1–5) — planning: action plan, WBS, topic framing, and this proposal.
- Phase 2 (Day 6–11) — prototype: Kafka plus streaming, model trained, dashboard, prototype showing.
- Phase 3 (Day 12–20) — refinement: benchmarks, fault-tolerance tests, tuning, final report and presentation. WBS reviewed on Day 9, 14 and 19.